Day 2: Life tables, YLL, and Arriaga

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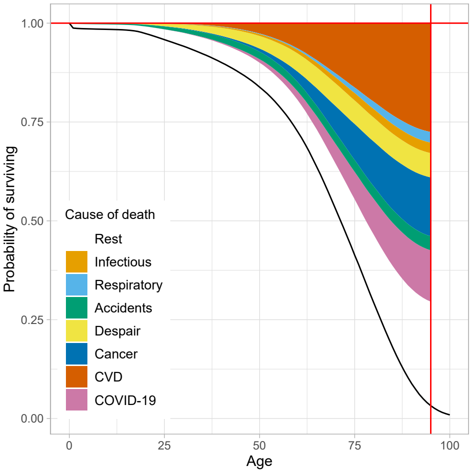
Kitagawa (1955)

To avoid the interaction term, Kitagawa suggests

$$\Delta CDR = \underbrace{\sum_x \left(\frac{M_x(t_2)+M_x(t_1)}{2}\right)\left(\frac{N_x(t_2)}{N(t_2)}-\frac{N_x(t_1)}{N(t_1)}\right)}_{\text{Changes in x-composition}} + \underbrace{\sum_x \left(\frac{\frac{N_x(t_2)}{N(t_2)}+\frac{N_x(t_1)}{N(t_1)}}{2}\right)(M_x(t_2)-M_x(t_1))}_{\text{Changes in rates}} \quad (1)$$

Life years lost: Andersen et al (2013)



Source: Aburto et al 2022

Outline

- Difference in life expectancy

Life expectancy decomposition

In the 1980s, several authors developed similar approaches: Pollard (1982), Andreev (1982), Arriaga (1984), Pressat (1985) ...

We focus on Arriaga's method

Arriaga (1984)

Effects of mortality change by age groups on life expectancies ($\sum_n \Delta_x =$ Total change):

$$ {}_n\Delta_x = \underbrace{\frac{\ell_x^1}{\ell_0^1}\left(\frac{{}_nL_x^2}{\ell_x^2} - \frac{{}_nL_x^1}{\ell_x^1}\right)}_{\text{Direct effect}} + \underbrace{\frac{T_{x+n}^2}{\ell_0^1}\left(\frac{\ell_x^1}{\ell_x^2} - \frac{\ell_{x+n}^1}{\ell_{x+n}^2}\right)}_{\text{Indirect and interaction effects}} $$

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- Change in mortality rates between ages x and $x+n \longrightarrow$ effect of change in number of years lived between x and $x+n$ on life expectancy.

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Note: for the open-ended age group there is only direct effect:
${}_\infty\Delta_x = \frac{\ell_x^1}{\ell_0^1}\left(\frac{T_x^2}{\ell_x^2} - \frac{T_x^1}{\ell_x^1}\right)$

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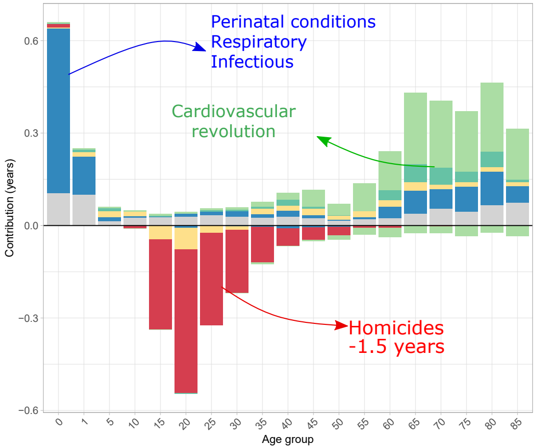
\begin{aligned}
{}_n\Delta_x^i &= {}_n \Delta_x \frac{{}_n m_{x,i}^2 - {}_n m_{x,i}^1}{{}_n m_x^2 - {}_n m_x^1} \\
&= {}_n \Delta_x \frac{{}_n R_{x,i}^2 [{}_n m_x^2] - {}_n R_{x,i}^1 [{}_n m_x^1]}{{}_n m_x^2 - {}_n m_x^1}
\end{aligned}
$$

where ${}_n R_{x,i}^t$ is the proportion of deaths from cause i in age groups x to $x + n$ in population t, and ${}_n\Delta_x$ is the contribution of all-cause mortality differences in the same age group. ${}_n m_x^t$ is the mortality rate in the same age groups of population t.

Venezuela: Change in male life expectancy at birth by cause of death 1996-2013, from 68.6 to 70.6



García & Aburto 2019. International Journal of Epidemiology

Arriaga's method

Exercise 2